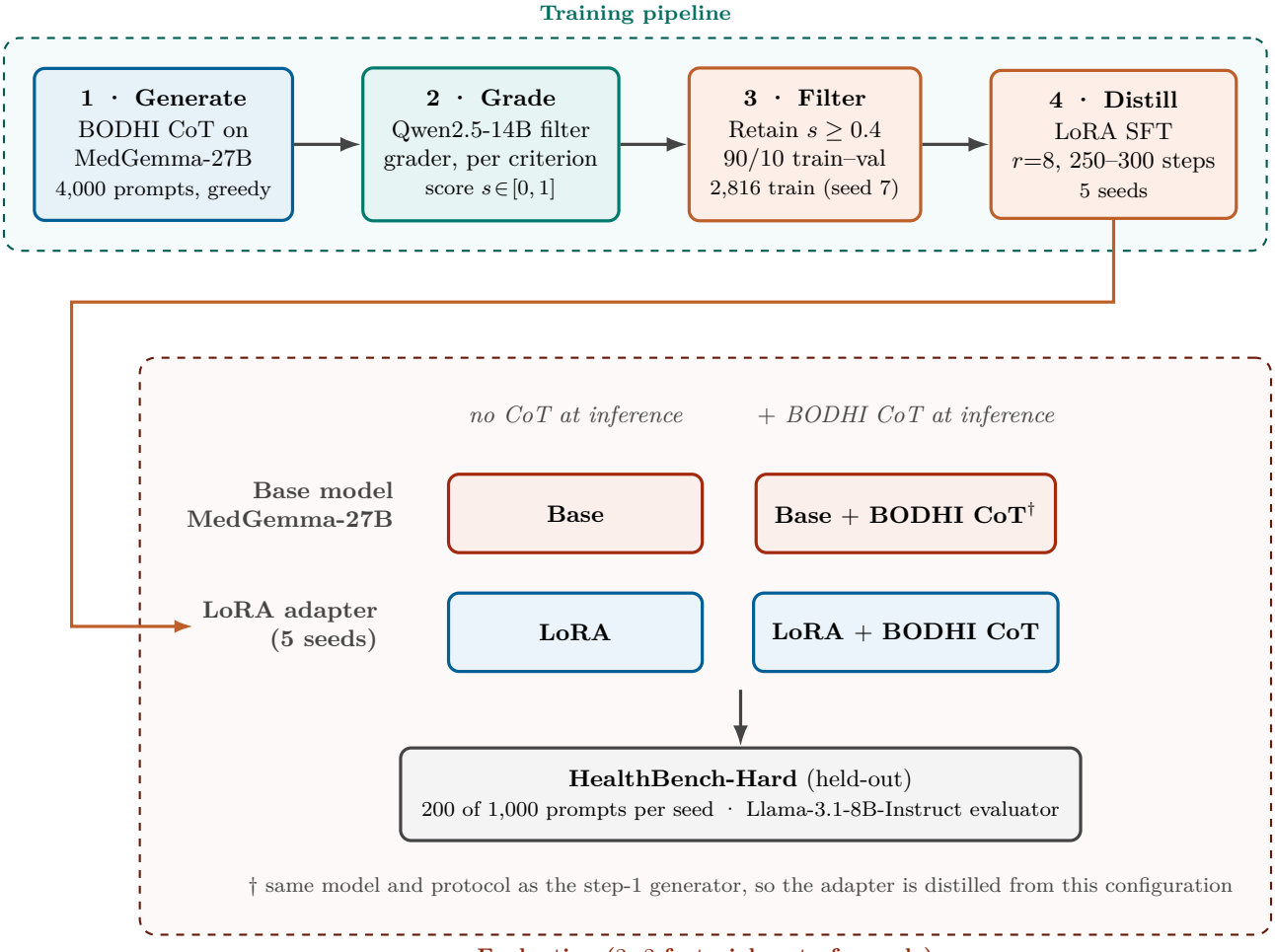
Training pipeline
1 · Generate
BODHI CoT on MedGemma-27B
4,000 prompts, greedy
2 · Grade
Qwen2.5-14B filter grader, per criterion score $s \in [0, 1]$
3 · Filter
Retain $s \geq 0.4$
90/10 train–val
2,816 train (seed 7)
4 · Distill
LoRA SFT
r=8, 250–300 steps
5 seeds
no CoT at inference
+ BODHI CoT at inference
Base model MedGemma-27B
Base
Base + BODHI CoT†
LoRA adapter (5 seeds)
LoRA
LoRA + BODHI CoT
HealthBench-Hard (held-out)
200 of 1,000 prompts per seed · Llama-3.1-8B-Instruct evaluator
† same model and protocol as the step-1 generator, so the adapter is distilled from this configuration
Evaluation (2×2 factorial, out-of-sample)